

HackerRank 3rd Semester Portfolio

Algorithmic Problem-Solving Activity

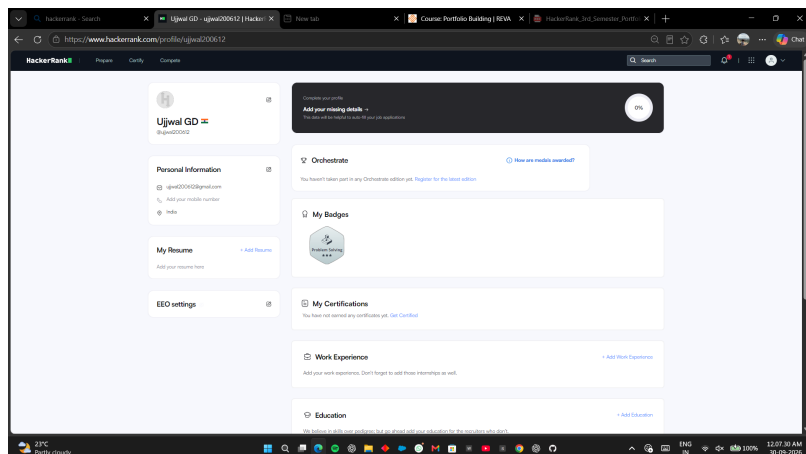
Student	Ujjwal GD
GitHub Repository	https://github.com/ujjwal200612-sketch/HackerRank-3rdSem-Portfolio
HackerRank Profile	https://www.hackerrank.com/profile/ujjwal200612

Problems Completed

No.	Problem	Main Concept	Time	Space
1	Diagonal Difference	2D Arrays	$O(N)$	$O(1)$
2	Dynamic Array	Vectors, XOR	$O(N + Q)$	$O(N)$
3	Time Conversion	Strings and Logic	$O(1)$	$O(1)$
4	Compare the Triplets	Comparison	$O(1)$	$O(1)$
5	Sparse Arrays	Hash Map / Frequency Counting	$O(N + Q)$	$O(N)$

Submission Evidence

HackerRank 3-star Problem Solving badge:



HackerRank Profile and Badge Evidence

The profile screenshot shows my public HackerRank profile and my 3-star Problem Solving badge.

[Open HackerRank Profile](https://www.hackerrank.com/profile/ujjwal200612)

Reflection on Optimization Techniques

During this activity, I learned how choosing a suitable algorithm can make a program simpler and more efficient. While solving the HackerRank problems, I practiced array traversal, string processing, vectors, XOR operations, frequency counting, and basic comparisons. I also started paying more attention to time and space complexity instead of only checking whether the output was correct.

For problems such as Diagonal Difference and Compare the Triplets, a single pass through the required data was enough, giving constant extra space. Dynamic Array helped me understand how vectors and XOR can be used together to organize and access data efficiently. Sparse Arrays showed the advantage of using a hash map for frequency counting instead of repeatedly scanning the complete list of strings. Time Conversion was mainly about handling conditions and string positions carefully.

I also practiced additional HackerRank problems while working toward the Problem Solving badge. These exercises helped me become more comfortable with loops, arrays, conditions, and writing functions that fit an existing program template. Overall, the activity improved my ability to break a problem into smaller steps, choose an appropriate approach, and think about efficiency before writing the final code.

Key Learning Outcomes

- Improved understanding of algorithmic problem solving.
- Practiced evaluating time and space complexity.
- Used vectors, hash maps, arrays, strings, and XOR operations.
- Improved familiarity with HackerRank workflows.
- Built a GitHub portfolio containing accepted solutions and evidence.

Portfolio Links

GitHub:

<https://github.com/uijwal200612-sketch/HackerRank-3rdSem-Portfolio>

HackerRank:

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